

Bayesian Model Selection for Heavy-Tailed Data

CLATSE 2025 Ecuador

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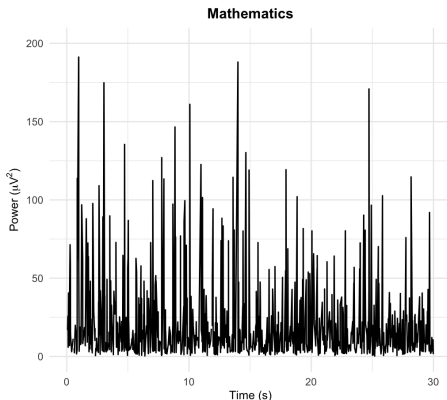
October 21, 2025

Presentation Overview

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- 4 Simulation Study
- 5 Applications
- 6 Discuss

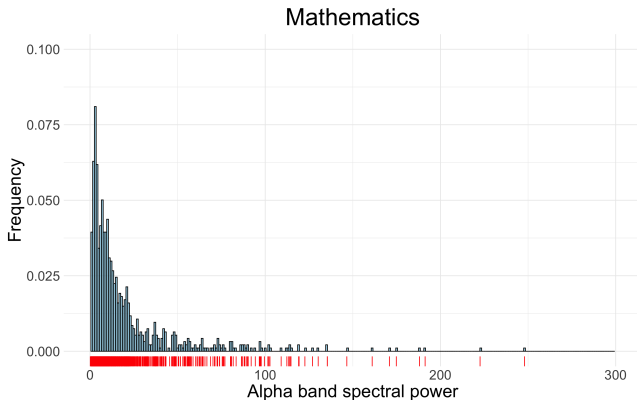
EEG signals analysis study from UC Berkeley

- The EEG signal tracks the activity of neurons in frequency bands.
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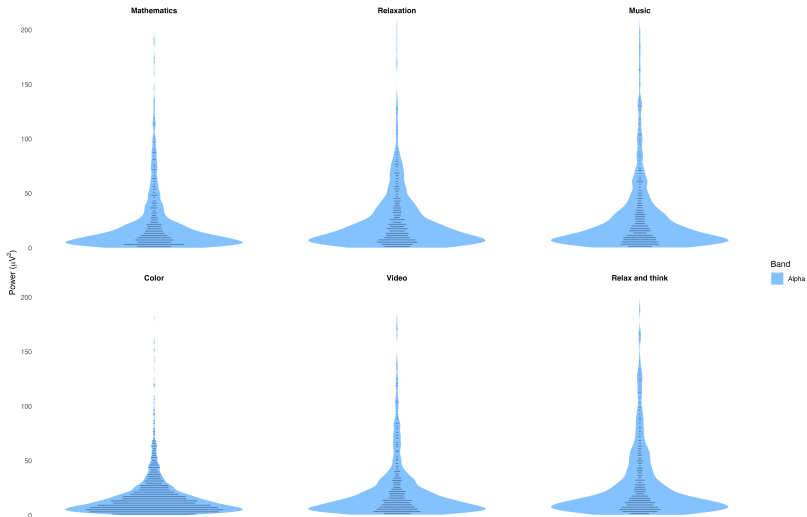


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- The EEG signal tracks the activity of neurons in frequency bands.
- This study involved 30 participants who were subject to several audio-visual stimuli. ($n = 7,506$)
- Activity in the alpha and beta bands follows a heavy-tailed distribution (Roberts et al., 2015).



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 - Generalized extreme value (GEV) distribution

$$F_{GEV}(y; \mu, \sigma, \xi) = \exp\left\{-\left[1 + \xi\left(\frac{y - \mu}{\sigma}\right)\right]^{-1/\xi}\right\}, \quad 1 + \xi\left(\frac{y - \mu}{\sigma}\right) > 0,$$

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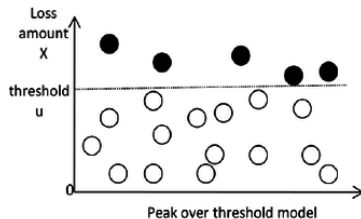
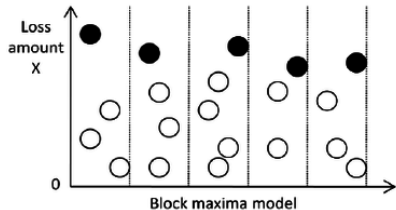
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 - Block Maxima (BM) (Gumbel, 1958)
 - Peak-over-Threshold (PoT) (Pickands, 1975) → Generalized Pareto Distribution (GPD)

$$F_{GPD}(y; \mu, \sigma, \xi) = 1 - \left[1 + \xi \left(\frac{y - \mu}{\sigma} \right) \right]^{-1/\xi}, \quad y \geq \mu, \quad 1 + \xi \left(\frac{y - \mu}{\sigma} \right) > 0,$$

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- Model selection to explain what information explains heavy-tailed data.

The Problem

Let $Y \in \mathbb{R}_+$ the response variable associated with predictors $\mathbf{x} \in \mathbb{R}^p$.
In particular, let $\{\mathbf{x}^j : j \in 2^p\}$ be set of all a subset of $\mathbf{x}$

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$$Y \mid \mathbf{x} \stackrel{\text{ind}}{\sim} F_{Y|\mathbf{x}}$$

$$F_{Y|\mathbf{x}} \sim \mathcal{RV}_\alpha$$

$$F_{Y|\mathbf{x}} = F_{Y|\mathbf{x}^j}$$

$$\mathbf{x}^j \subseteq \mathbf{x}$$

Proposal Model

Let $\mathbf{y} = (y_1, \dots, y_n)^\top$ be random realizations from a heavy-tailed distribution with associated fixed predictors $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$. Model selection over a space defined by these predictors can be encoded via a binary vector $\gamma \in \{0, 1\}^p$. The j -th model can be obtained via the Hadamard product $\mathbf{x}^j = \mathbf{x} \odot \gamma^j$.

$$\begin{aligned} y_i \mid \mathbf{x}_i, \gamma^j, F_{Y|\mathbf{x} \odot \gamma^j} &\sim F_{Y|\mathbf{x} \odot \gamma^j}, \quad i = 1, \dots, n; j = 1, \dots, 2^p. \\ F_{Y|\mathbf{x} \odot \gamma^j} &\sim \Pi_1(F_{Y|\mathbf{x} \odot \gamma^j}), \\ \gamma^j &\sim \Pi_2(\gamma^j). \end{aligned} \quad (1)$$

Heavy-Tailed Distributions

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Definition: Heavy-Tailed Distributions

A distribution Y is heavy-tailed if its survival function $S_Y(y)$ satisfies:

$$\limsup_{y \rightarrow \infty} e^{\lambda y} S_Y(y) = \infty, \forall \lambda > 0.$$

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Definition: Regular Variation $\mathcal{RV}_\alpha$

A distribution is $\mathcal{RV}_\alpha$ if:

$$\lim_{y \rightarrow \infty} \frac{S_Y(yt)}{S_Y(y)} = t^{-\alpha}, \alpha > 0.$$

Heavy-Tailed Distributions

Example 1: Generalized Pareto Distribution

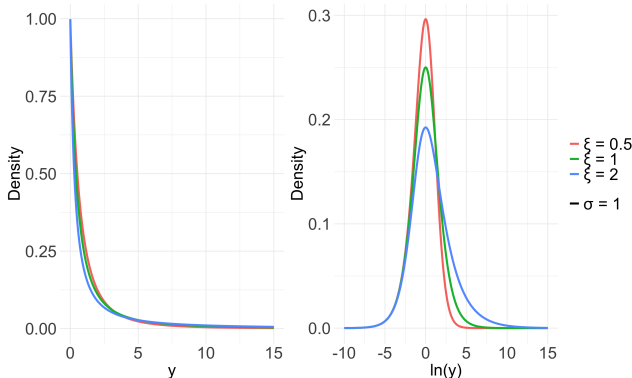
$Y \sim \text{GPD}(\sigma, \xi)$. The support of Y is $[0, \infty)$ if $\xi \geq 0$, and $[0, -\sigma/\xi)$ if $\xi < 0$.
The distribution of Y is: $F_Y(y) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}$,

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Log Transformation Heavy-Tailed Distribution

Theorem

Let $Y \in \mathbb{R}_+$ be a random variable following a regularly varying distribution F_Y with index $\alpha > 0$. Let $Z = \ln(Y)$ be a transformation. Then, the random variable Z follows a light-tailed distribution F_Z .

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$$z_i = \ln(y_i) \mid \mathbf{x}_i, \gamma^j, F_{Z|\mathbf{x} \odot \gamma^j} \sim F_{Z|\mathbf{x} \odot \gamma^j}, \quad i = 1, \dots, n, \quad j = 1, \dots, J, \quad (2)$$

$$F_{Z|\mathbf{x} \odot \gamma^j} \sim \Pi_1^*(F_{Z|\mathbf{x} \odot \gamma^j}). \quad (3)$$

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$$F_{Y|\mathbf{x}} = \begin{cases} F_{Z|\mathbf{x}}(\ln y), & \text{if } y > 0, \\ 0, & \text{if } y \leq 0. \end{cases} \quad (4)$$

Model in Logarithmic Space

Model the conditional distribution of Z given the predictors $\mathbf{x} \odot \boldsymbol{\gamma}^j$, which exhibits light-tail behavior.

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$$z_i \mid \mathbf{x}_i, \gamma^j, \mathbf{s}_i, \sim \mathcal{N}(x_i^t \beta_{s_i}, \sigma_{s_i}^2), \quad i = 1, \dots, n, \quad j = 1, \dots, J,$$

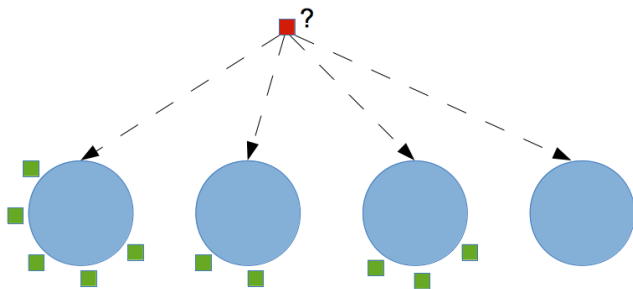
$$(\mathbf{s}_1, \dots, \mathbf{s}_n) \sim \text{CRP}(M),$$

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Chinese Restaurant representation (CRP) (Teh et al., 2006)



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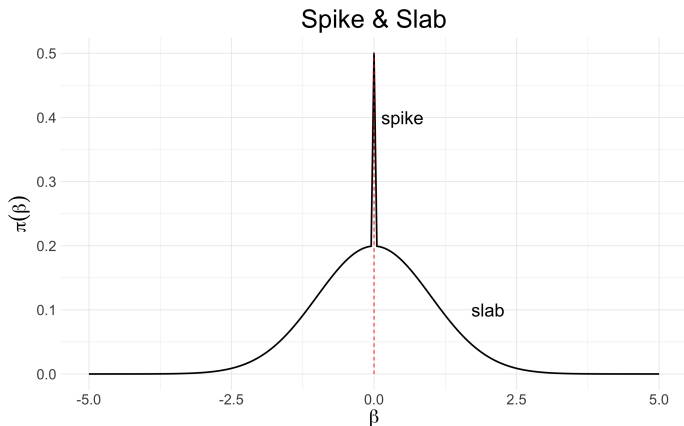
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Spike-and-Slab Prior (Mitchell and Beauchamp, 1988)



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Prior over model space

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Bernoulli Prior(George and McCulloch, 1993)

$$p(\gamma \mid \omega) = \prod_{j=1}^p \omega^{\gamma_j} (1 - \omega)^{1 - \gamma_j}$$

where ω is a hyperparameter.

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Beta-Binomial Prior(Ley and Steel, 2009)

$W = \sum_{i=1}^p \gamma_i \sim \text{Binomial}(p, \omega)$ and has a probability mass function given by

$$P(W = w) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)\Gamma(a+b+p)} \binom{p}{w} \Gamma(a+w)\Gamma(b+p-w), \quad w = 0, \dots, p.$$

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Prior over model space

Womack et al. (2015)

$$p(\gamma) = \zeta_0 \sum_{\gamma': \gamma \preceq \gamma'} p(\gamma'),$$

where $\gamma \preceq \gamma'$ if and only if $\gamma' \neq \gamma$ and $\gamma_i \leq \gamma'_j$ for all $1 \leq i, j \leq p$, $\zeta_0 \in (0, \infty)$ is a hyperparameter.

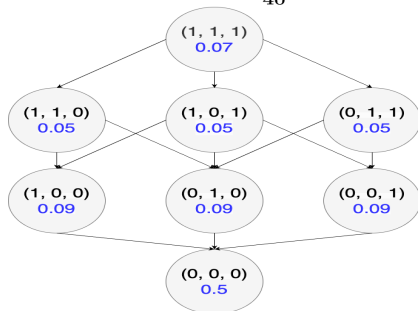
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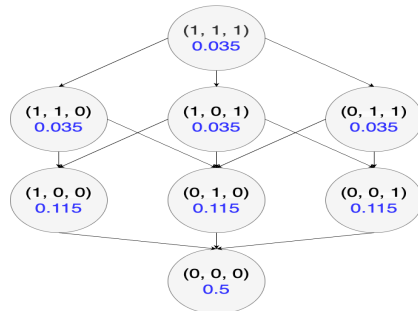
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Beta-Binomial ($a = \frac{29}{40}, b = 2$)



Womack ($\zeta = 1$)



Stochastic search in model space MCMC

$$p(\gamma \mid y, X) \propto f(y \mid \gamma, X)p(\gamma)$$

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On Metropolis-Hastings Step, by indicating with γ^{new} the candidate model, the acceptance probability is calculated as

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How select the new candidate?

Proposal Model

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$$(\gamma_1, \dots, \gamma_p) \sim \text{Womack}(\zeta)$$

Posterior Inference

Algorithm 1: Gibbs sampler.

- 1: Initialize $\gamma^{(0)}$, $M^{(0)}$ and $\mathbf{c}^{(0)}$.
- 2: Draw $\mathbf{c}_i^{(r)}$, $i = 1, \dots, n$ from $p(\mathbf{c}_i | \mathbf{c}_{-i}, \mathbf{z}, \mathbf{x}, \gamma)$ using Algorithm 3 in Neal (2000).
- 3: Draw $M^{(r)}$ from $p(M | \mathbf{c}^{(r-1)})$ using the method in Escobar and West (1995).
- 4: Draw $\gamma^{(r)}$ from $p(\gamma | \mathbf{z}, \mathbf{c}^{(r-1)})$ using a Metropolis-Hasting step with acceptance probability

$$\min \left(1, \frac{p(\gamma^{\text{new}} | \mathbf{y}, \mathbf{c}^{(r-1)})}{p(\gamma^{\text{old}} | \mathbf{y}, \mathbf{c}^{(r-1)})} \right) = \min \left(1, \frac{p(\mathbf{z} | \gamma^{\text{new}}, \mathbf{x})p(\gamma^{\text{new}})}{p(\mathbf{z} | \gamma^{\text{old}}, \mathbf{X})p(\gamma^{\text{old}})} \right),$$

- 5: Repeat steps 2–4 until reaching stationarity.
-

Posterior Inference

- Existing cluster:

$$\begin{aligned} p(c_i = c \mid c_{-i}, z, x, \gamma) &= b \cdot \frac{n_{-i,c}}{n-1+\alpha} \int_{\phi} f(z_i \mid \phi, \mathbf{x}_i \odot \gamma) \cdot dH_{-i,c}(\phi) \\ &= b \cdot \frac{n_{-i,c}}{n-1+\alpha} T(z_i \mid \mu_{-i,c,\gamma}, \Sigma_{-i,c,\gamma}, \nu_{-i,c,\gamma}) \end{aligned}$$

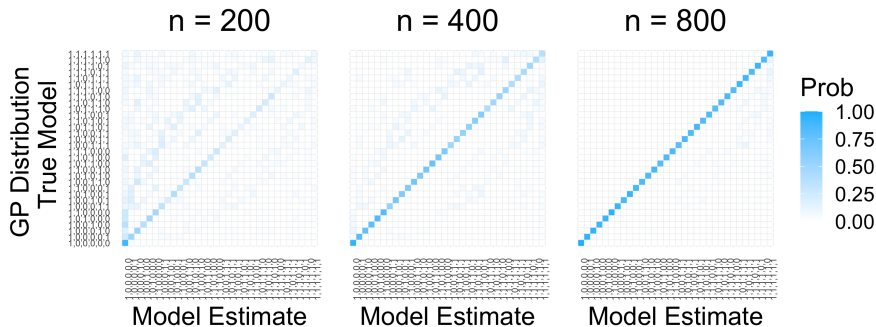
- New cluster:

$$\begin{aligned} p(c_i \neq c_j \forall j \neq i \mid c_{-i}, Z, X, \gamma) &= b \cdot \frac{\alpha}{n-1+\alpha} \int_{\phi} f(z_i \mid \phi, X_i^{\Gamma_1}) \cdot d_0(\phi) d\phi \\ &= b \cdot \frac{\alpha}{n-1+\alpha} T(z_i \mid \mu_{i,\gamma}, \Sigma_{i,\gamma}, \nu_{i,\gamma}) \end{aligned}$$

Simulation Study

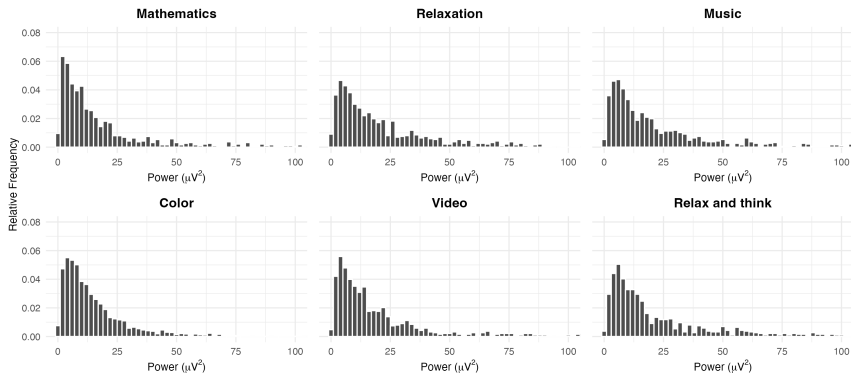
- Let $y_i \stackrel{iid}{\sim} GPD(\sigma = 1, \xi = \sqrt{\exp(x_i^\top \beta)})$, the tail index ($\alpha = 1/\xi$).
- Baseline scenario in which the parameter vector is defined as $\beta = (1, -1, 1, -1, 1, -1)$. The first element represents the intercept term. The design matrix X was generated randomly.
- Regarding sample size, we analyzed datasets with $n = 200, 400, 800$ observations to assess the convergence rate of the estimated model toward the true data-generating process.
- we systematically explored the $2^5 = 32$ possible submodels arising from all combinations of active predictors influencing the response variable's tail index.
- Each scenario was replicated $n_{mc} = 100$ times through Monte Carlo simulations to assess the model's behavior robustly.

Simulation Study

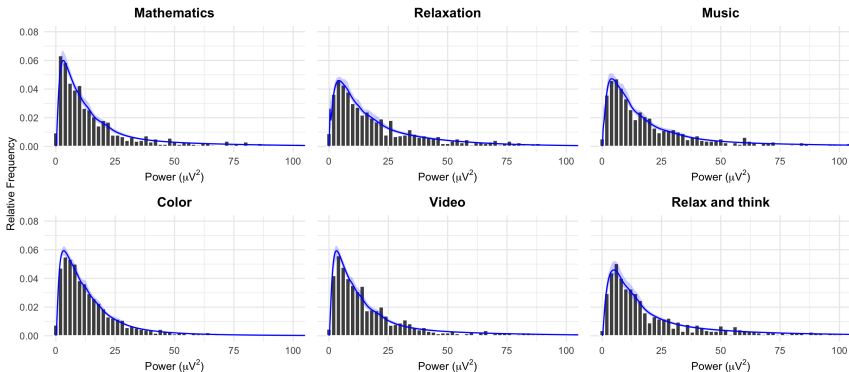


EEG signals analysis study from UC Berkeley (Alpha Band)

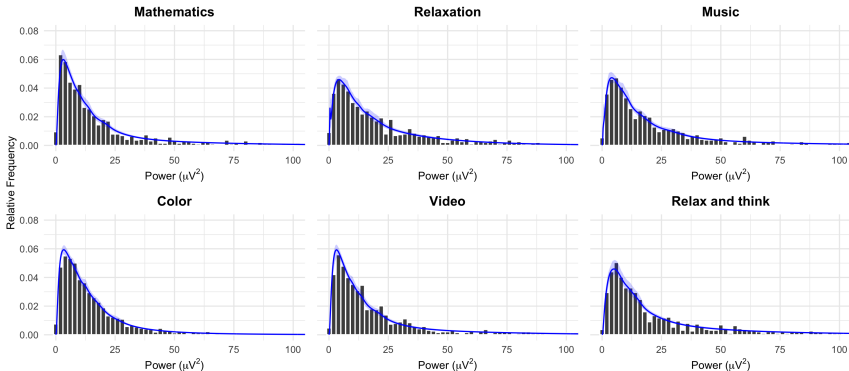
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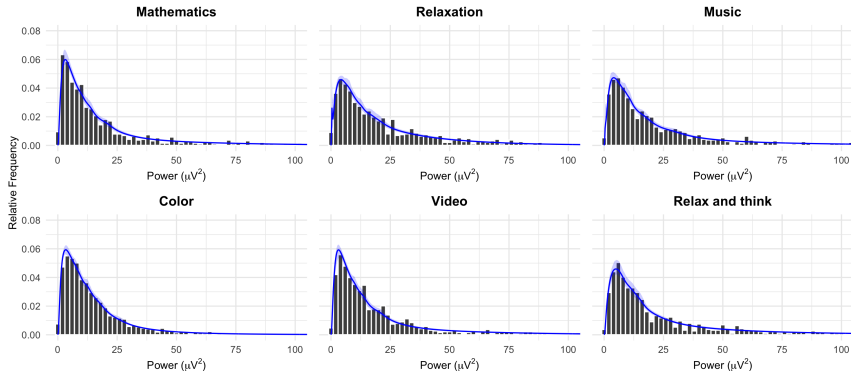


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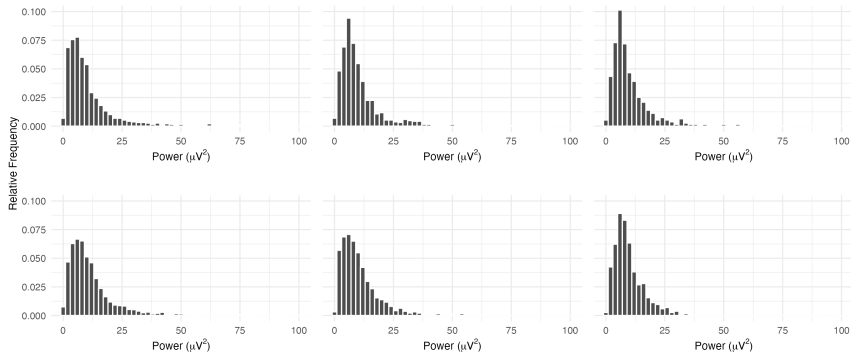
Stimulus	Relaxation	Music	Color	Video	Relax and think
Model Selection alpha band	0.999	0.994	1.000	0.559	0.999

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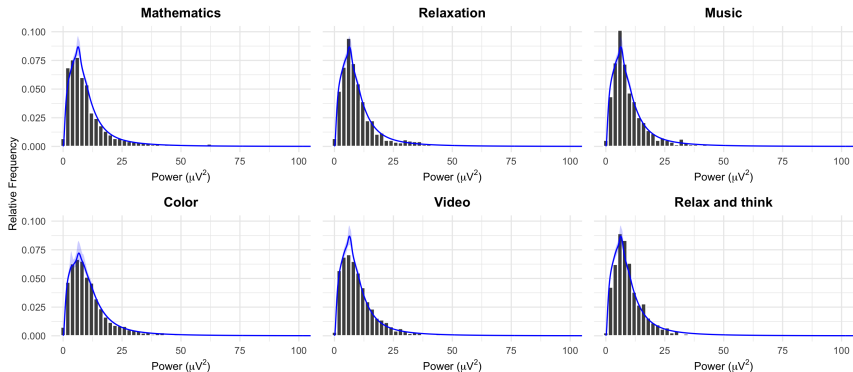


Alpha Band γ Posterior		
(1,1,1,1,1)	(1,1,1,0,1)	Others
0,560	0,434	0,006

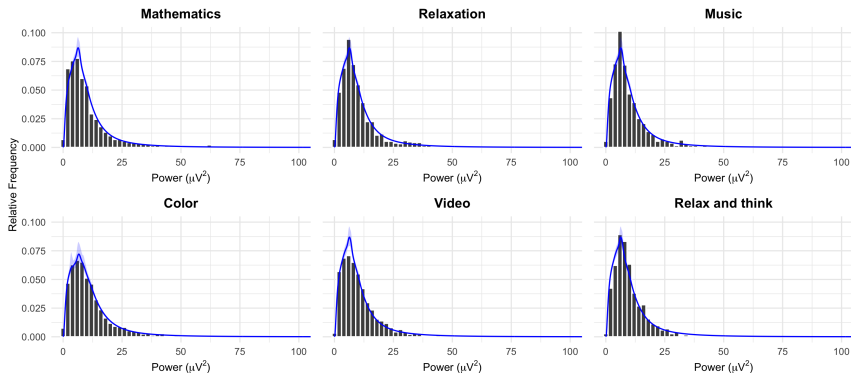
EEG signals analysis study from UC Berkeley (Beta Band)



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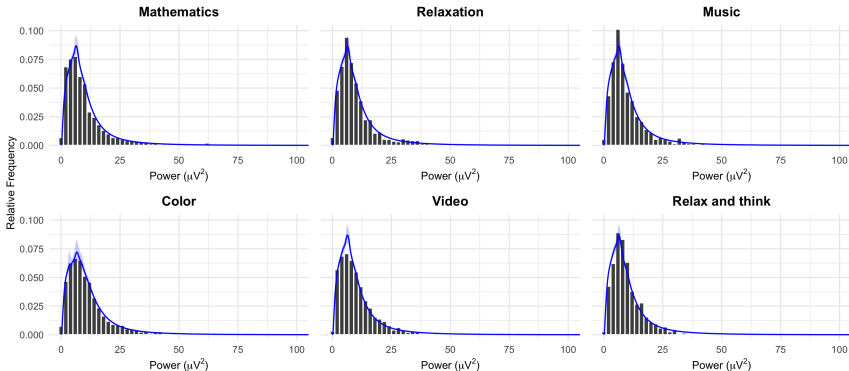


EEG signals analysis study from UC Berkeley (Beta Band)



Stimulus	Relaxation	Music	Color	Video	Relax and think
Model Selection beta band	0.000	0.000	1.000	0.007	0.000

EEG signals analysis study from UC Berkeley (Beta Band)



Beta Band γ Posterior	
(0,0,1,0,0)	(0,0,1,1,0)
0,993	0,007

Discuss

- Why mixture model
- Consistency $p(y_{\text{new}} | \mathbf{y}) \xrightarrow{n \rightarrow \infty} f_0(y_{\text{new}})$
- Is it possible other kernel in the mixture?
- How influence the prior in model space?

Thanks¹

¹Daniel Saavedra was funded by the National Agency for Research and Development (ANID)/Scholarship Program/Doctorado Nacional/2023-21231037.

Questions? Comments?

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